DATA 259 Peer Review Session 2 Document

Group: ethical-group (Max Linton, Alvin He, Charlie) **Topic:** BNPL Social Media Targeting: An Ethical Analysis

Edited Abstract

Buy Now, Pay Later (BNPL) services are rapidly expanding, yet their advertising practices remain largely unexamined under a data ethics framework. This project investigates the primary research question: *Do BNPL providers disproportionately target financially vulnerable audiences in their social media advertising—and if so, does the content of those ads obscure or downplay the risks of borrowing?* Leveraging the Meta Ad Library API, we collect ad-level data for six major BNPL providers, including age/gender targeting parameters, ad spend ranges, and ad creative text. We contextualize these targeting patterns using demographic financial fragility indicators from the U.S. Census American Community Survey (ACS). Furthermore, we employ Natural Language Processing (NLP) techniques to analyze the ad copy, evaluating readability, sentiment, and the use of risk-related versus urgency-inducing language across different demographic targets. By examining both who sees the ads and what the ads say, this research addresses a critical gap in consumer protection—the "legal but harmful" marketing of financial products to those most vulnerable to the algorithmic amplification of inequality. Our findings aim to provide a data-driven foundation to evaluate whether BNPL advertising constitutes predatory design.

Rough Draft of a New Section: Methods

Data Collection and Processing Pipeline Our methodology relies on a two-pronged data collection strategy to evaluate both targeting behavior and audience vulnerability. First, we programmatically query the Meta Ad Library API to collect comprehensive ad-level data from six major BNPL providers (Klarna, Affirm, Afterpay, Zip, Sezzle, and PayPal Pay Later). This yields structured data on age and gender targeting ranges, estimated audience sizes, ad spend brackets, and the raw ad creative text. Second, we gather population-level financial fragility indicators—such as debt-to-income ratios and emergency savings rates—from the U.S. Census American Community Survey (ACS) using the Census API. This allows us to map ad targeting parameters against demographic vulnerability.

For the analytical phase, we employ chi-square and proportion tests to compare age targeting distributions against the general adult population. Finally, our Natural Language Processing (NLP) pipeline analyzes the ad copy to test for systematic differences in messaging strategy. We utilize Flesch-Kincaid scores for readability, VADER and TextBlob for sentiment analysis, and term frequency analysis to contrast the prevalence of risk-downplaying language with aspirational or urgency-inducing terminology across different targeted groups.

What Our Group Has Already Done

To answer our project question, our group has successfully established the project repository and data pipeline infrastructure. We have written and executed automated scripts (`meta_api_pull.py` and `census_pull.py`) to systematically extract advertising data from the Meta Ad Library API and contextual demographic data from the U.S. Census API. These raw datasets have been ingested and structured in our initial Jupyter notebooks.

Furthermore, we have built the foundational NLP pipeline (`nlp_analysis.py`) to compute readability scores and sentiment polarity on the ad creatives. We have also started running preliminary statistical tests in our targeting and spend analysis notebooks to identify initial discrepancies in how different age brackets are targeted by these providers.

Long-Term Plans for Answering the Project Question

Our long-term plans focus on integrating these separate data streams into a cohesive ethical analysis and preparing for the final poster presentation. We will perform intersectional analyses to see if specific combinations of demographics face disproportionate ad spend or exposure to high-urgency messaging. We also intend to finalize the NLP keyword analysis to quantitatively map the prevalence of risk-obscuring language across different targeted demographics. Ultimately, these quantitative findings will be framed within our core ethical themes—autonomy, algorithmic amplification of inequality, and the "legal but harmful" paradigm—to draw concrete conclusions about predatory design in BNPL marketing, which will form the visual and narrative core of our final deliverable and poster.

Areas for Specific Feedback

We would appreciate specific feedback from our peer critique partners on the following areas:

1. **NLP Approaches:** Are there additional NLP metrics or specific sentiment dictionaries we should consider using to better capture "urgency" or "risk-downplaying" language in very short ad creatives?
2. **Visualizations:** How can we most effectively visualize the intersection of Census vulnerability data with Meta's broad age/gender targeting ranges on a poster format without overwhelming the audience?
3. **Ethical Framing:** Does our framing of the "legal but harmful" ethical concept clearly connect to our quantitative methodology, or does it require a more explicit operational definition in our final report?